Kernel: Python 3 (Anaconda)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
df = pd.read_csv('education_data.csv')
education_data_df = pd.read_csv("education_data.csv")
education_data_df.head()
```

Out[4]:

	id	school_name	state	school_type	grade_level	funding_per_student_usd	avg_
0	1	Bowman High School	Michigan	Private	High	9575.80	65.9
1	2	Foster High School	Michigan	Charter	Middle	16733.10	87.3
2	3	Henson High School	New York	Public	Middle	24890.74	72.8
3	4	Weaver High School	Texas	Private	Elementary	6857.49	81.5
4	5	King Elementary School	Michigan	Public	Elementary	16910.73	44.7

```
education_data_df["state"].value_counts()
```

Out[5]:

```
state
Texas             119
Florida           110
Georgia           107
Illinois          102
Ohio              101
California         97
North Carolina     96
New York           95
Michigan           89
Pennsylvania       84
Name: count, dtype: int64
```

In [8]:

```
print(education_data_df.columns)
```

Out[8]:

```
Index(['id', 'school_name', 'state', 'school_type', 'grade_level',
       'funding_per_student_usd', 'avg_test_score_percent',
       'student_teacher_ratio', 'percent_low_income', 'percent_minority',
       'internet_access_percent', 'dropout_rate_percent'],
      dtype='object')
```

In [10]:

```
print(education_data_df["state"].unique())
```

Out[10]:

```
['Michigan' 'New York' 'Texas' 'Pennsylvania' 'Georgia' 'Illinois'
 'California' 'North Carolina' 'Ohio' 'Florida']
```

```
¡ TASK 01 :: INDIVIDUAL VISUALIZATIONS
Create at least two of the following visualizations:
  Histogram or boxplot for a numerical column (e.g., income, score, age).
```

```
¡ Bar chart for a categorical column (e.g., region, department, gender).
¢ Count plot showing frequency distribution of a key variable.
  EXPLAIN: What does this visualization reveal about the distribution or
spread of values?
```

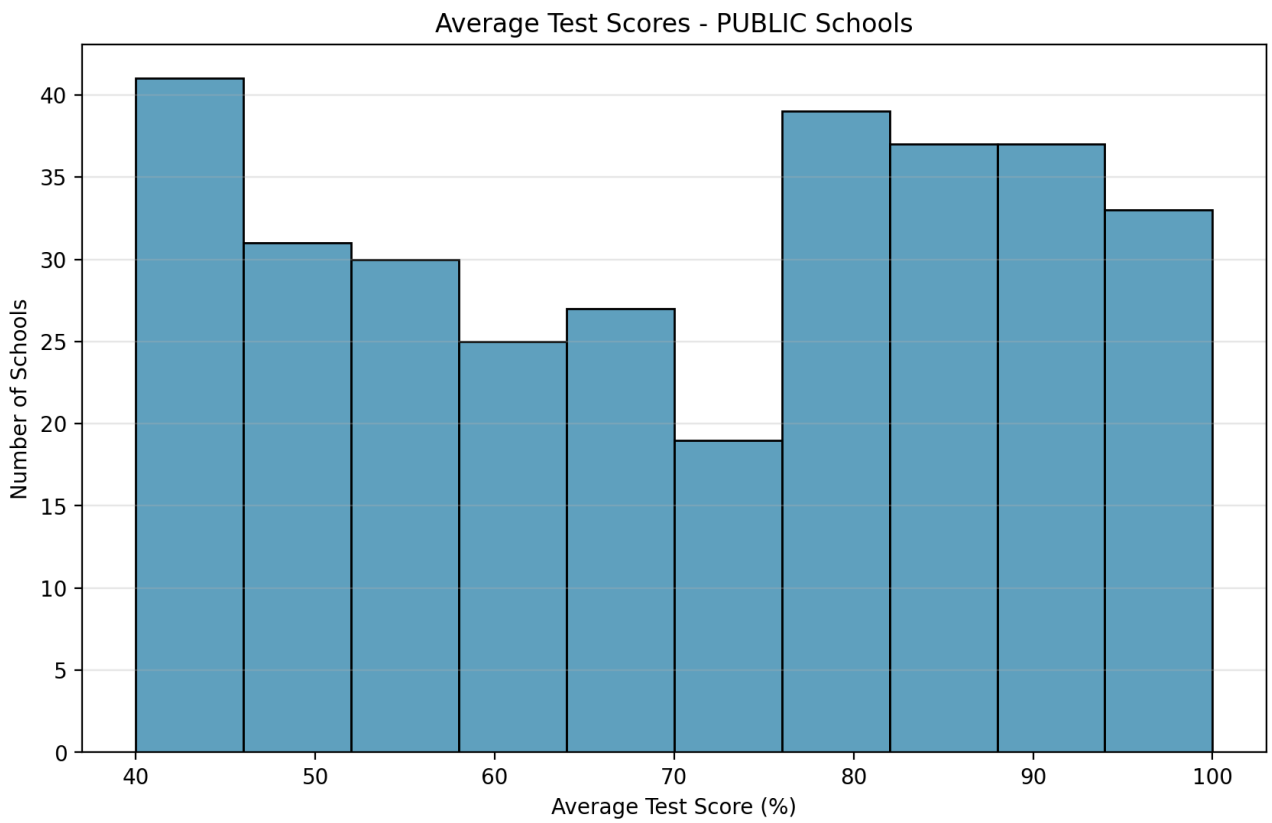
Out[7]:

```
  Cell In[7], line 1
    ¡ TASK 01 :: INDIVIDUAL VISUALIZATIONS
    ^
SyntaxError: invalid character '¡' (U+00A1)
```

In [15]:

```
#Histogram of Public Shools

public_schools = education_data_df[education_data_df['school_type'] ==
'Public']

plt.figure(figsize=(10, 6))
sns.histplot(data=public_schools,
             x='avg_test_score_percent',
             bins=10,
             color='#2E86AB',  # Blue
             edgecolor='black',
             stat='count')

plt.xlabel('Average Test Score (%)')
plt.ylabel('Number of Schools')
plt.title('Average Test Scores - PUBLIC Schools')
plt.grid(axis='y', alpha=0.3)
```

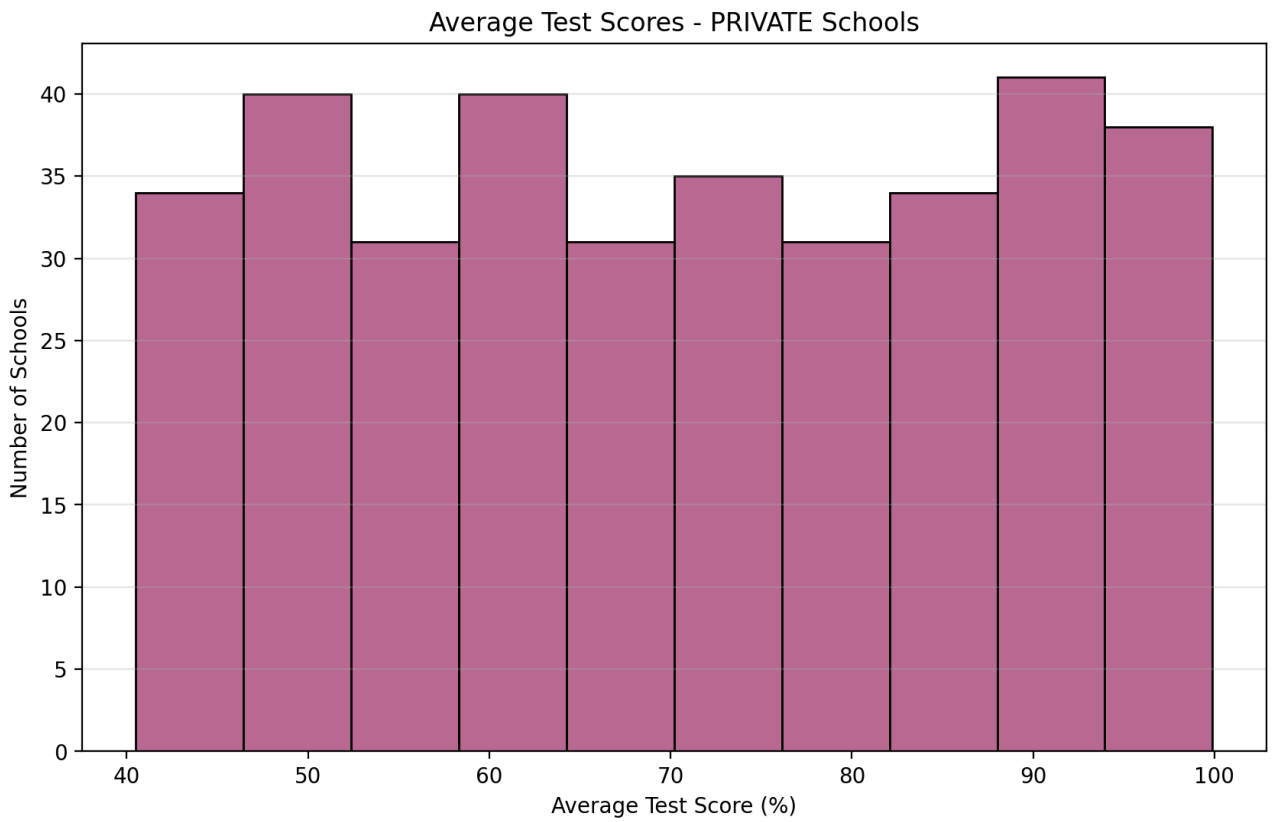
Out[15]:




```
# Hisotgram of Private School

private_schools = education_data_df[education_data_df['school_type'] ==
'Private']
```

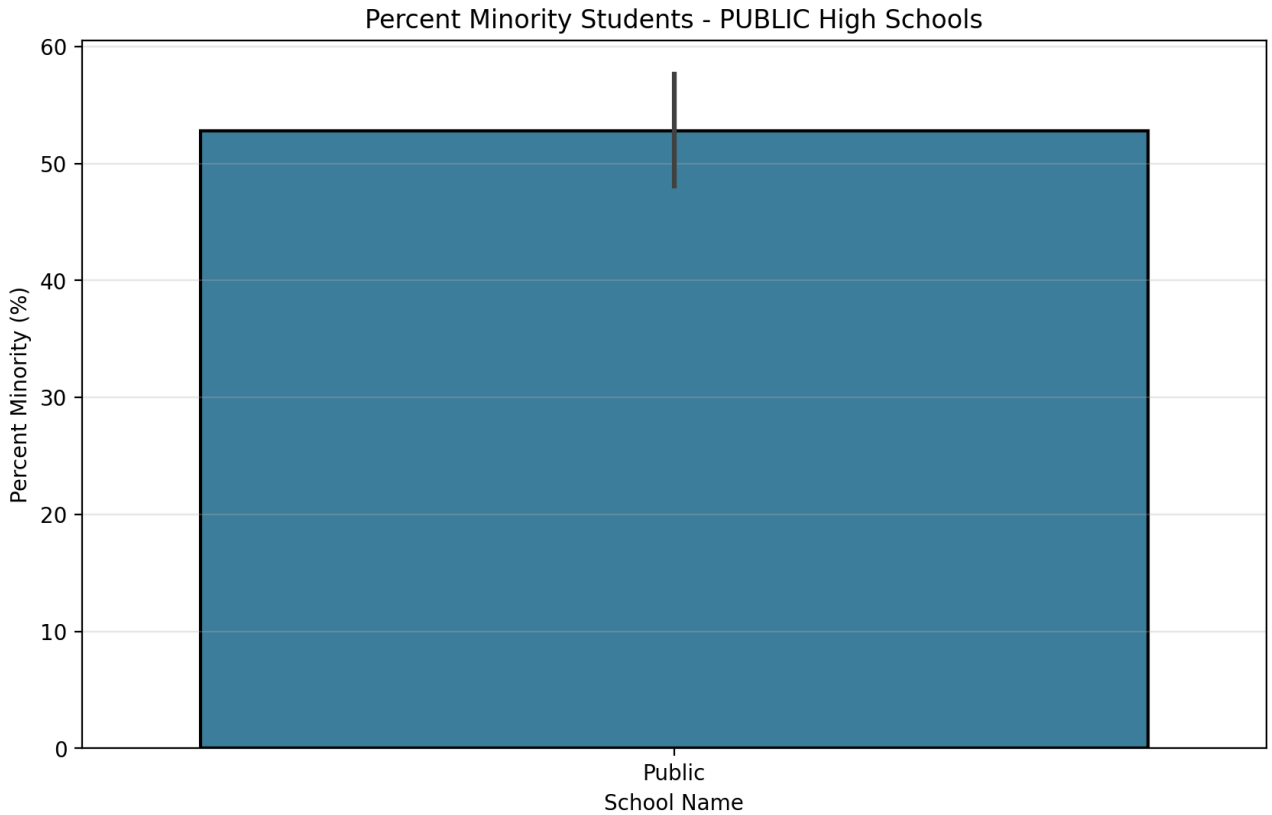
```
plt.figure(figsize=(10, 6))
sns.histplot(data=private_schools,
             x='avg_test_score_percent',
             bins=10,
             color='#A23B72',  # Pink/Magenta
             edgecolor='black',
             stat='count')

plt.xlabel('Average Test Score (%)')
plt.ylabel('Number of Schools')
plt.title('Average Test Scores - PRIVATE Schools')
plt.grid(axis='y', alpha=0.3)
```

Out[16]:




```
#Bar Chart


public_high = education_data_df[(education_data_df['school_type'] ==
'Public') &
                                (education_data_df['grade_level'] == 'High')]

plt.figure(figsize=(10, 6))
sns.barplot(data=public_high,
            x='school_type',
            y='percent_minority',
            color='#2E86AB',  # Blue
            edgecolor='black',
            linewidth=1.5)

plt.xlabel('School Name')
plt.ylabel('Percent Minority (%)')
plt.title('Percent Minority Students - PUBLIC High Schools')
plt.grid(axis='y', alpha=0.3)
```

Out[18]:

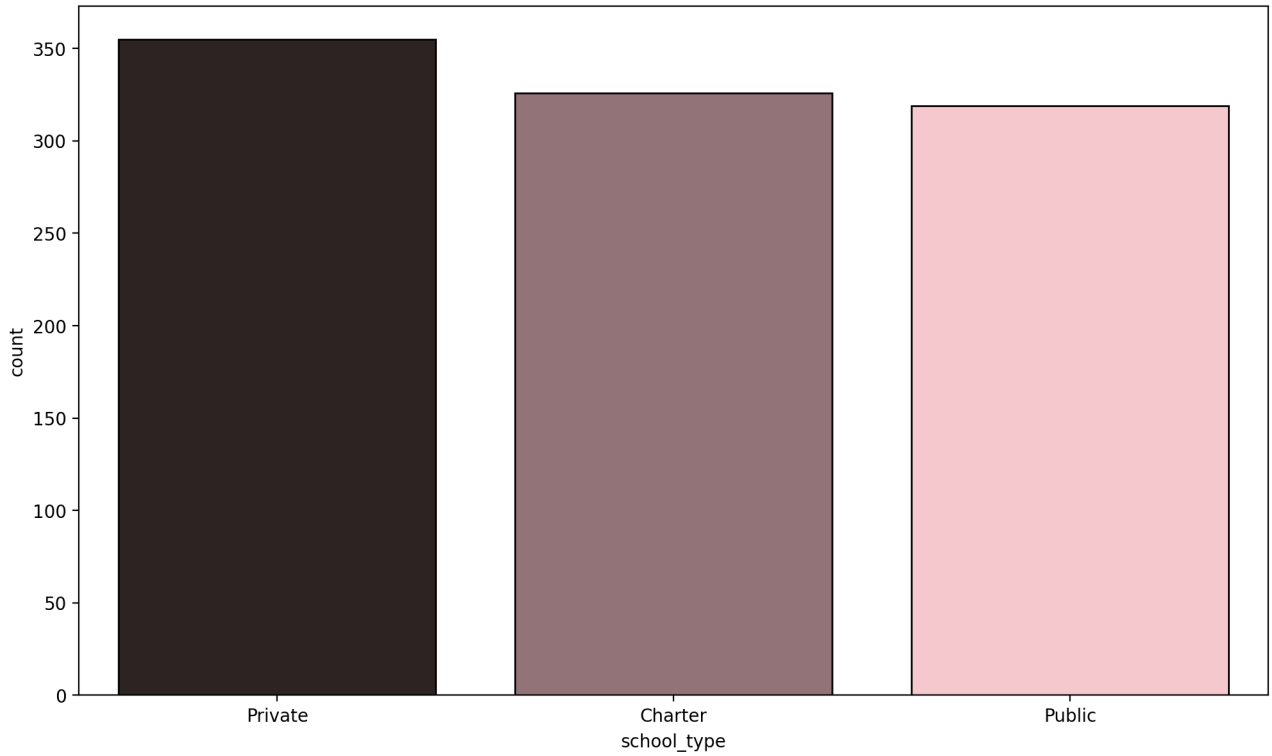


In [6]:

```
sns.countplot(data=education_data_df, x="school_type", color="pink",
edgecolor="black", hue = "school_type")
plt.show()
```

Out[6]:

```
/tmp/ipykernel_370/411085731.py:1: FutureWarning:

Setting a gradient palette using color= is deprecated and will be removed in
v0.14.0. Set `palette='dark:pink'` for the same effect.

  sns.countplot(data=education_data_df, x="school_type", color="pink",
edgecolor="black", hue = "school_type")
```




```
# Section 2: Relationship Visualizations
```

```
¨ RELATIONSHIP ANALYSIS
Use visualizations that reveal relationships between two numerical or
mixed variables:
i Scatter plot – Show relationship between two numerical variables.
j Heatmap of correlations – Visualize correlation matrix.
  EXPLAIN: Do you observe trends, correlations, or outliers?
3 sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

Out[10]:

```
  Cell In[10], line 1
    ¨ RELATIONSHIP ANALYSIS
    ^
SyntaxError: invalid character '¨' (U+00A8)
```

In [3]:

```
import matplotlib.pyplot as plt
import seaborn as sns
education_data_df
sns.cx="avg_test_score_percent", y="internet_access_percent", data =
education_data_df)
plt.title('Relationship Between Type of School and Percentage of Minority
Students')
plt.show()
```

Out[3]:

```
  Cell In[3], line 4
    sns.cx="avg_test_score_percent", y="internet_access_percent", data =
education_data_df)

^
SyntaxError: unmatched ')'
```

```
import pandas as pd
import matplotlib.pyplot as plt
```

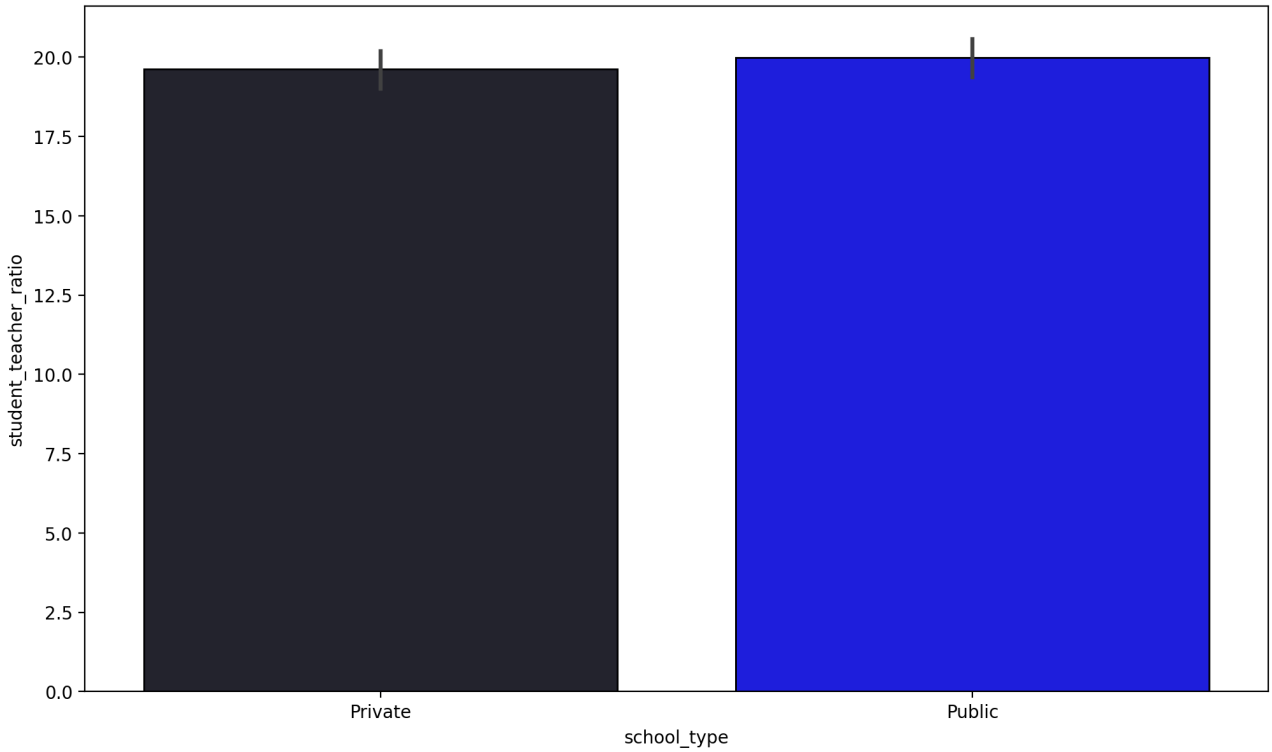
```
import seaborn as sns

df = education_data_df
print(df.shape)
df.head()
```

Out[56]: (1000, 12)

	id	school_name	state	school_type	grade_level	funding_per_student_usd	avg_
0	1	Bowman High School	Michigan	Private	High	9575.80	65.9
1	2	Foster High School	Michigan	Charter	Middle	16733.10	87.3
2	3	Henson High School	New York	Public	Middle	24890.74	72.8
3	4	Weaver High School	Texas	Private	Elementary	6857.49	81.5
4	5	King Elementary School	Michigan	Public	Elementary	16910.73	44.7

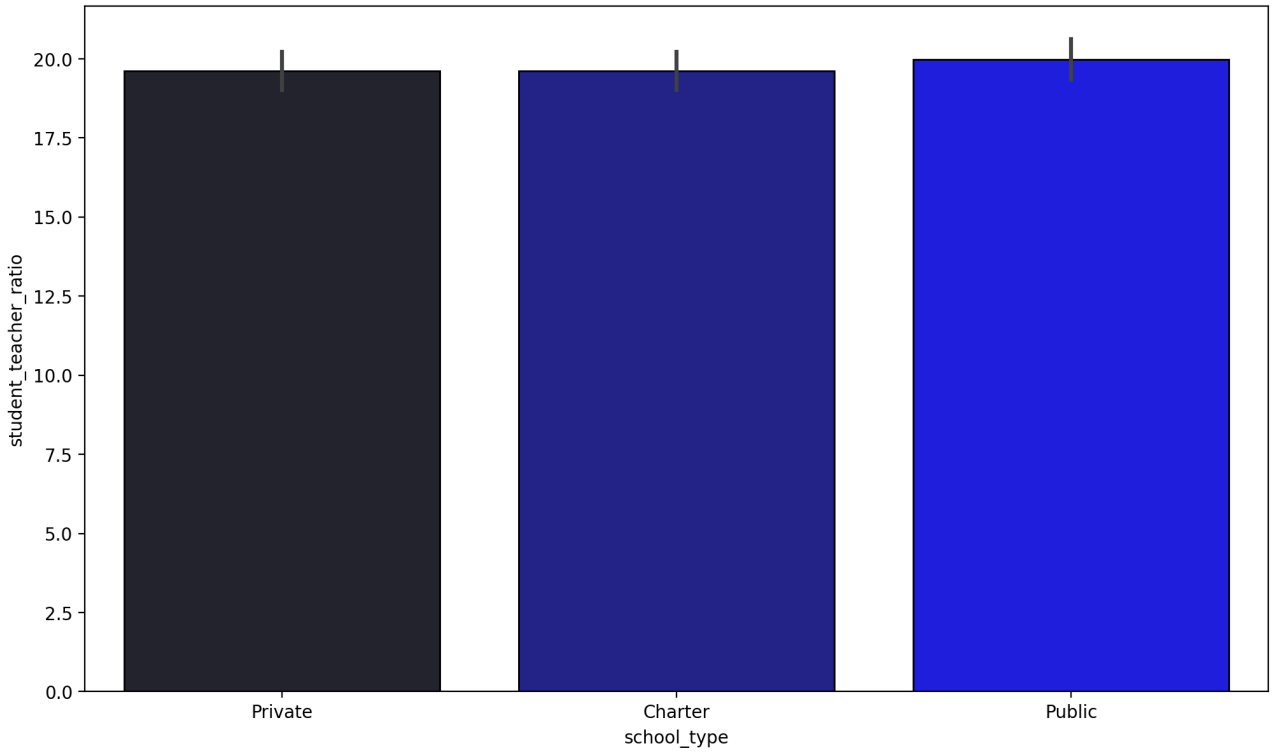
```
# Bar plot: average student_teacher_ratio by school_type
sns.barplot(

data=education_data_df[education_data_df["school_type"].isin(["Private",
"Public"])],
    x="school_type",
    y="student_teacher_ratio",
    color="blue",
    edgecolor="black",
    hue = "school_type"
)
plt.show()
```

Out[7]:
```
/tmp/ipykernel_370/1469619731.py:2: FutureWarning:

Setting a gradient palette using color= is deprecated and will be removed in
v0.14.0. Set `palette='dark:blue'` for the same effect.

  sns.barplot(
```



In [9]:
```
# Bar plot: average student_teacher_ratio by school_type
sns.barplot(
    data=education_data_df,
    x="school_type",
    y="student_teacher_ratio",
    color="blue",
    edgecolor="black",
    hue = "school_type"
)
plt.show()
```

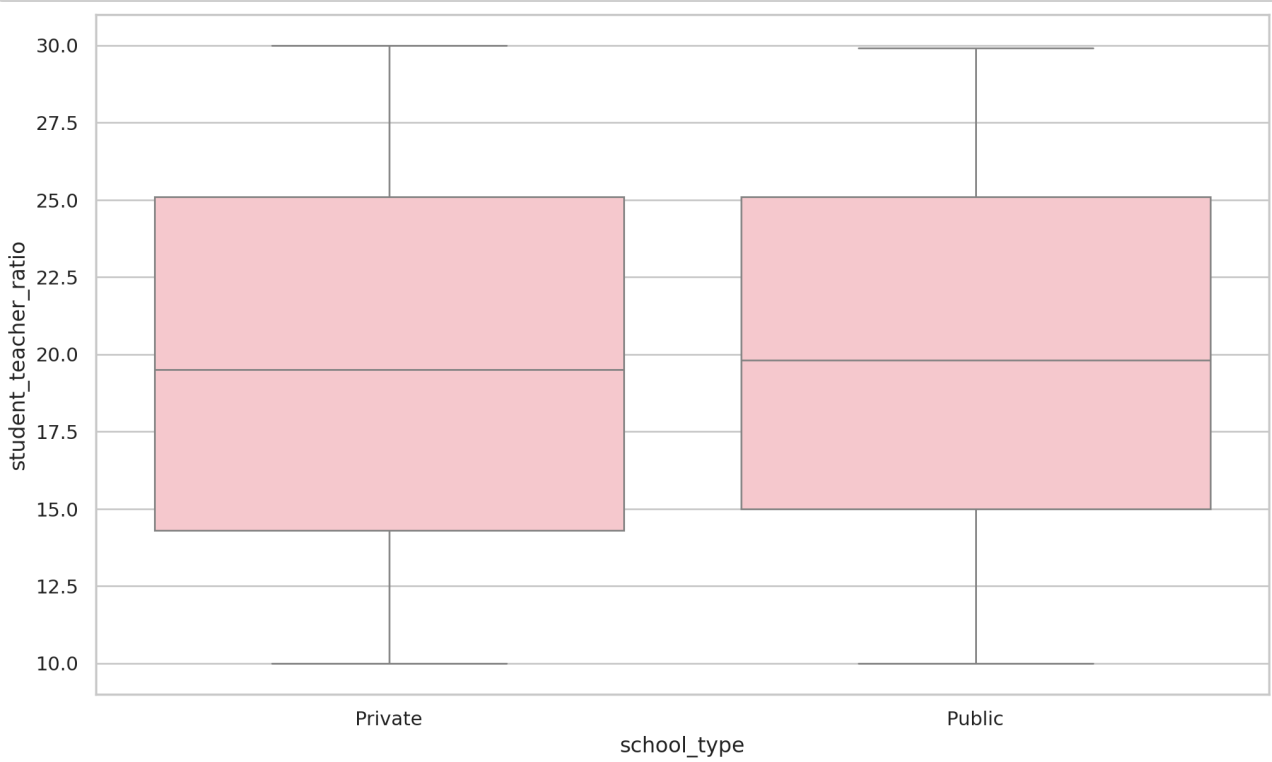
Out[9]:
```
/tmp/ipykernel_370/2759339308.py:2: FutureWarning:

Setting a gradient palette using color= is deprecated and will be removed in
v0.14.0. Set `palette='dark:blue'` for the same effect.

  sns.barplot(
```



In [85]:

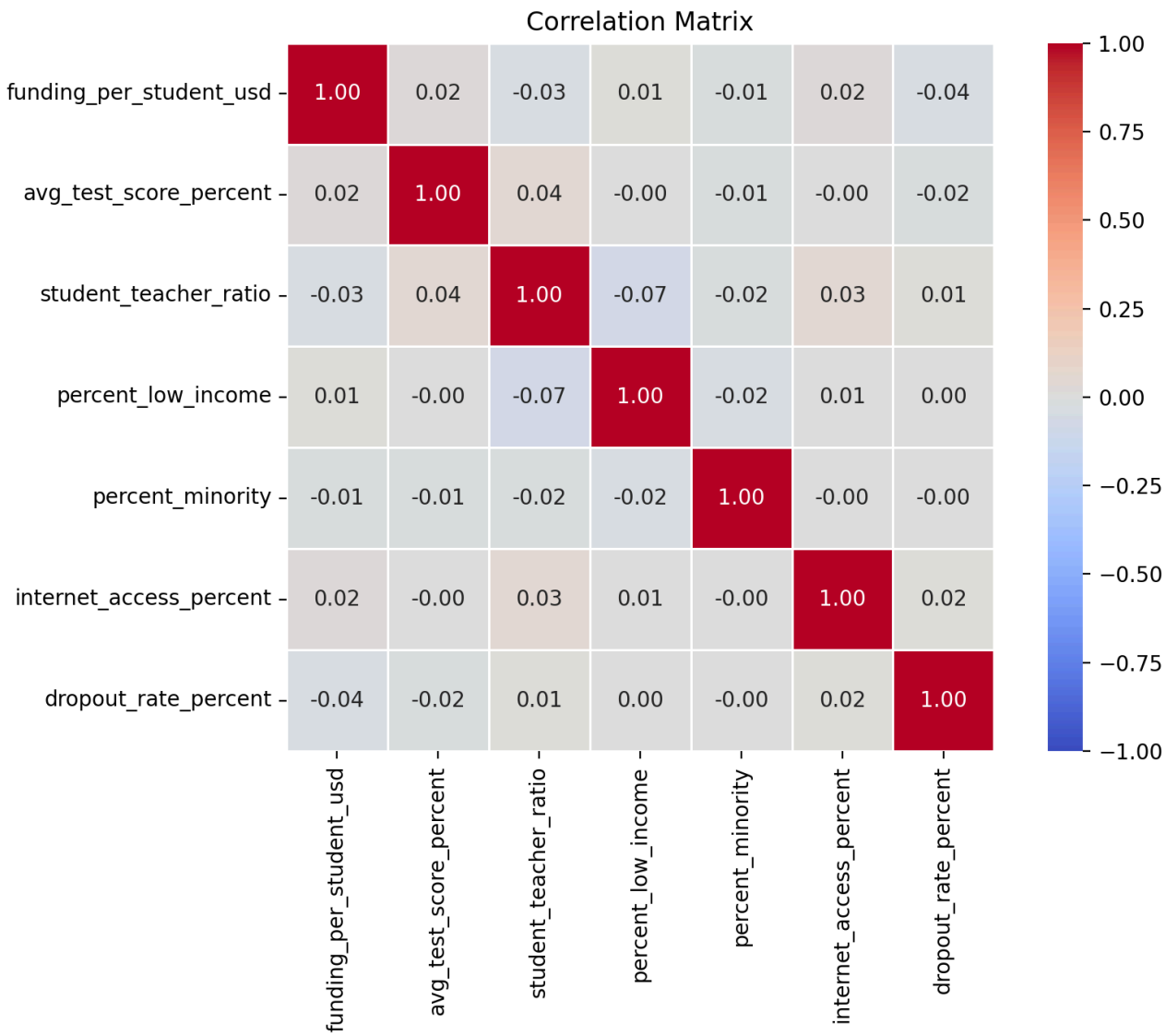
```
# Box plot: distribution of student_teacher_ratio by school_type
sns.boxplot(

data=education_data_df[education_data_df["school_type"].isin(["Private",
"Public"])],
    x="school_type",
    y="student_teacher_ratio",
    color="pink"
)
plt.show()
```

Out[85]:



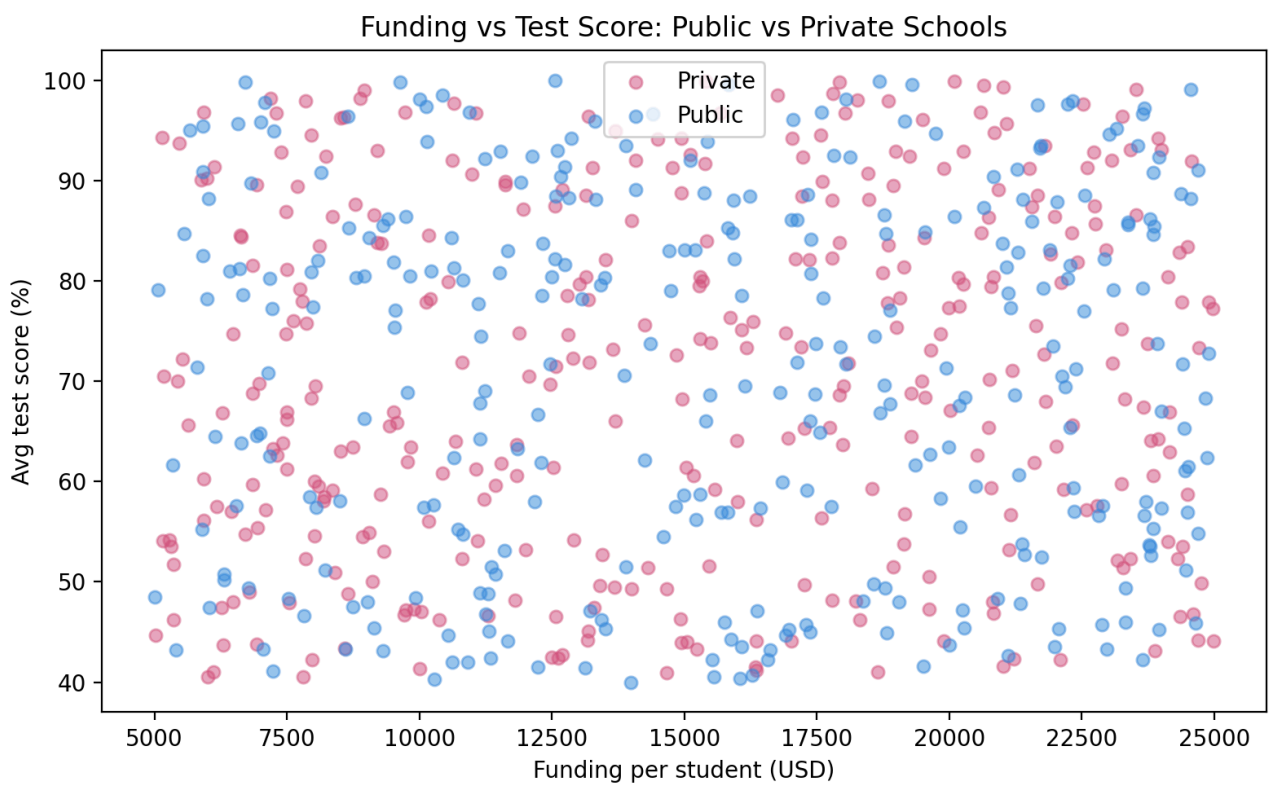
```
numeric_cols = [
    'funding_per_student_usd', 'avg_test_score_percent',
    'student_teacher_ratio', 'percent_low_income',
    'percent_minority', 'internet_access_percent',
    'dropout_rate_percent'
]

corr = df[numeric_cols].corr()

fig, ax = plt.subplots(figsize=(9, 7))
sns.heatmap(
    corr,
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    center=0,
    vmin=-1, vmax=1,
    square=True,
    linewidths=0.5,
    ax=ax
)
ax.set_title('Correlation Matrix')
plt.tight_layout()
plt.show()
```

Out[58]:



```
fig, ax = plt.subplots(figsize=(8, 5))

colors = {'Public': '#378ADD', 'Private': '#D4537E'}

filtered_df = df[df['school_type'].isin(['Public', 'Private'])]

for school_type, group in filtered_df.groupby('school_type'):
    ax.scatter(
        group['funding_per_student_usd'],
        group['avg_test_score_percent'],
        label=school_type,
        color=colors[school_type],
        alpha=0.5,
        s=30
    )

ax.set_xlabel('Funding per student (USD)')
ax.set_ylabel('Avg test score (%)')
ax.set_title('Funding vs Test Score: Public vs Private Schools')
ax.legend()
plt.tight_layout()
plt.show()
```

Out[59]:



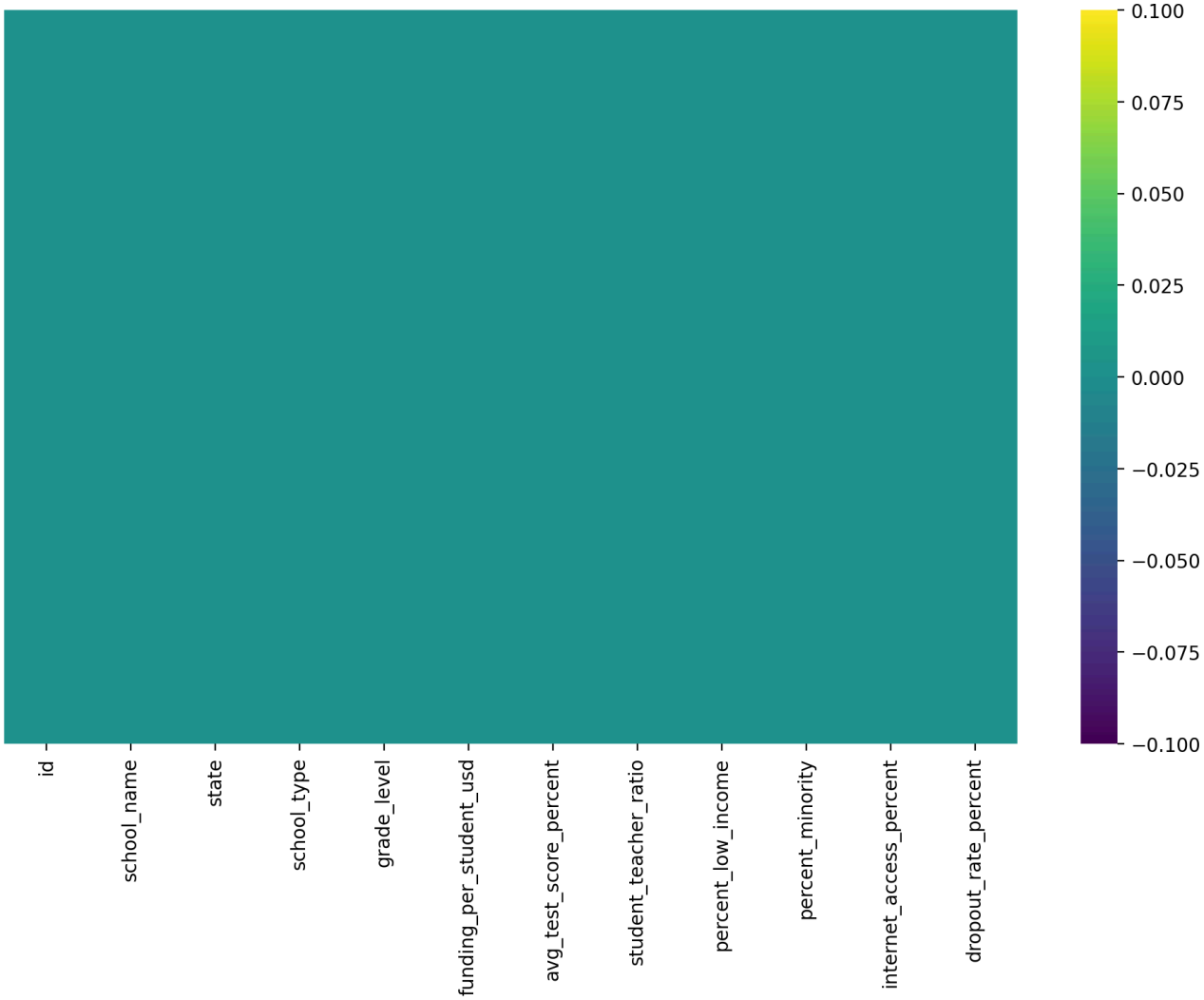

```
sns.scatterplot(x='student_teacher_ratio', data= education_data_df)
```

Out[4]:

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 1
----> 1 sns.scatterplot(x='student_teacher_ratio', data= education_data_df)
NameError: name 'sns' is not defined
```

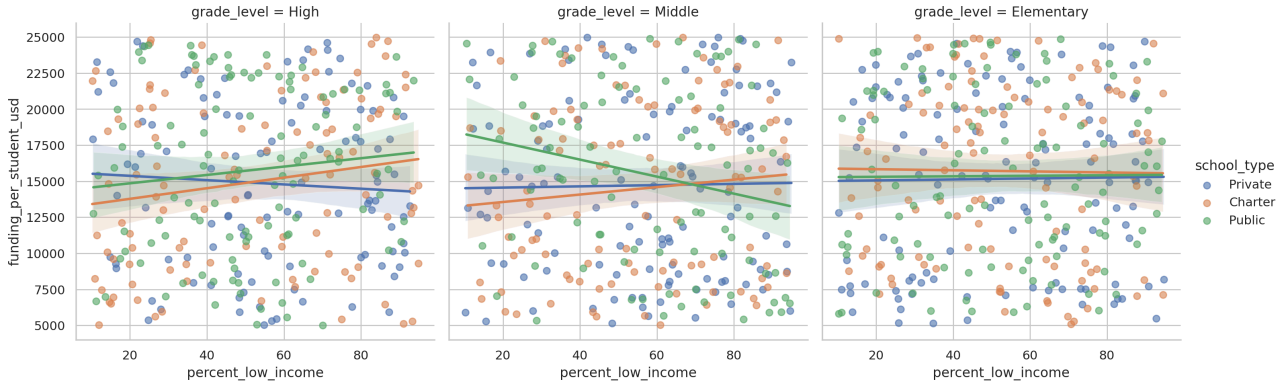
```
sns.heatmap(education_data_df.isnull(), yticklabels=False,
cmap='viridis')
```

Out[64]: <Axes: >




```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set(style="whitegrid")

sns.lmplot(
    data=df,
    x="percent_low_income",
    y="funding_per_student_usd",
    hue="school_type",
    col="grade_level",
    height=5,
    aspect=1,
    scatter_kws={"alpha": 0.6}
)

plt.show()
```

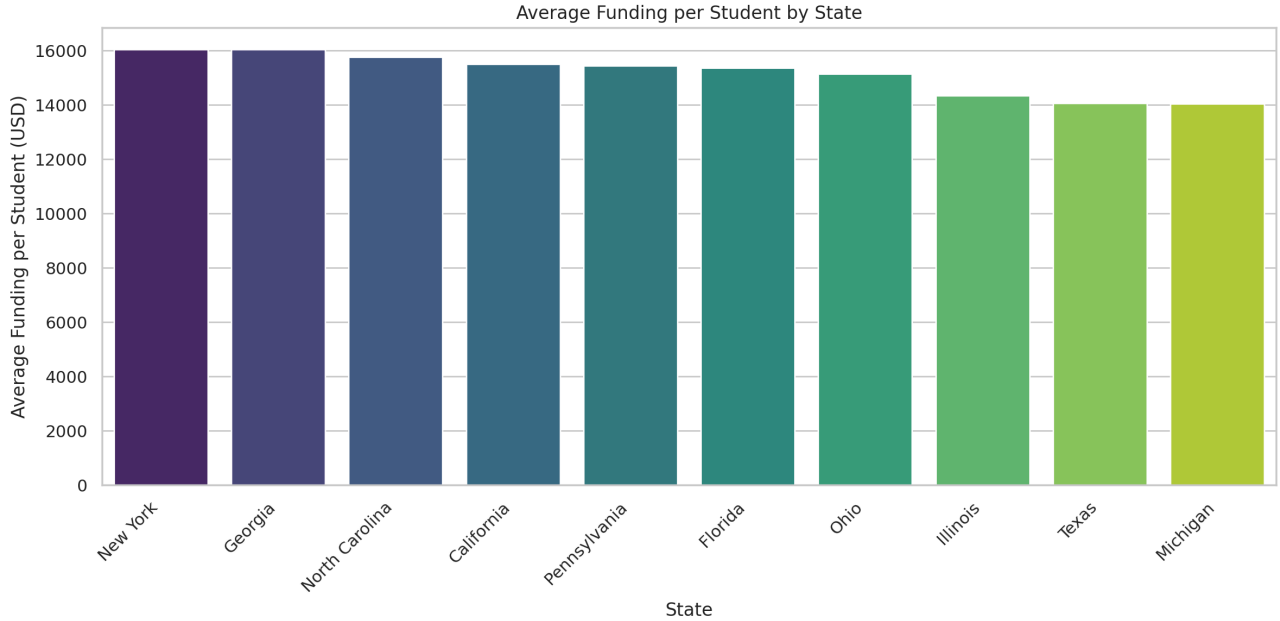
Out[86]:




```
state_funding = (
    df.groupby("state", as_index=False)["funding_per_student_usd"]
      .mean()
      .sort_values("funding_per_student_usd", ascending=False)
)

plt.figure(figsize=(12, 6))
sns.barplot(
    data=state_funding,
    x="state",
    y="funding_per_student_usd",
    palette="viridis"
)

plt.xticks(rotation=45, ha="right")
plt.xlabel("State")
plt.ylabel("Average Funding per Student (USD)")
plt.title("Average Funding per Student by State")
plt.tight_layout()
plt.show()
```

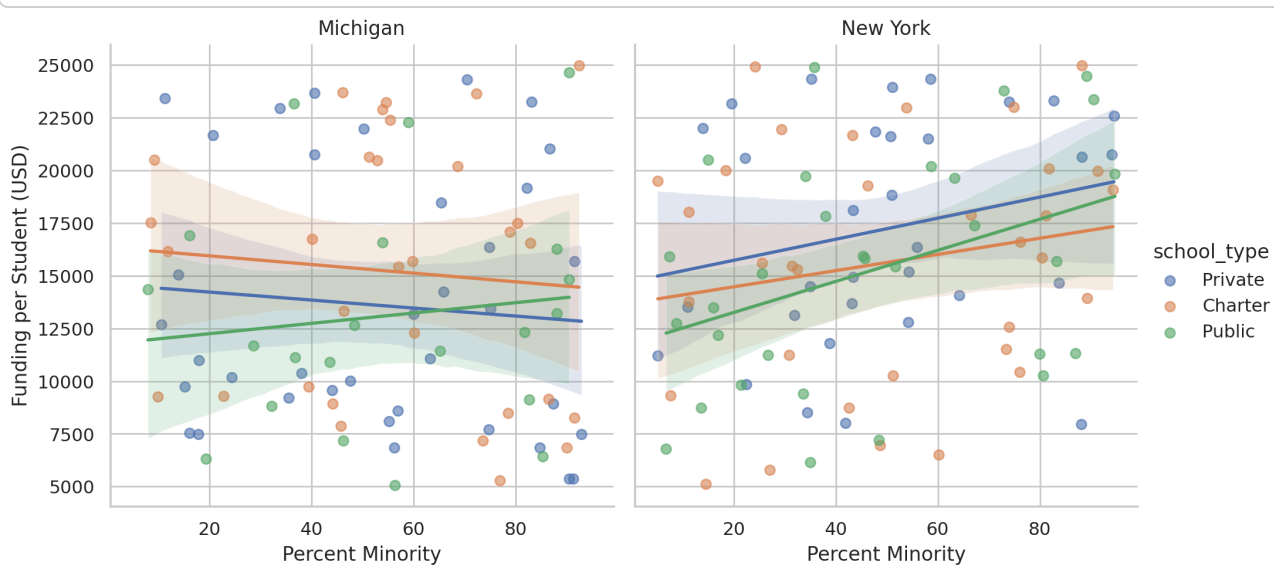
Out[90]:

```
/tmp/ipykernel_367/1314075558.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(
```




```
# Filter to the two states
ny_mi = df[df["state"].isin(["New York", "Michigan"])]

# Create side-by-side scatterplots with regression lines
g = sns.lmplot(
    data=ny_mi,
    x="percent_minority",
    y="funding_per_student_usd",
    hue="school_type",
    col="state",
    height=5,
    aspect=1,
    scatter_kws={"alpha": 0.6},
    line_kws={"linewidth": 2}
)

g.set_axis_labels("Percent Minority", "Funding per Student (USD)")
g.set_titles("{col_name}")
plt.show()
```

Out[91]:



In [ ]:

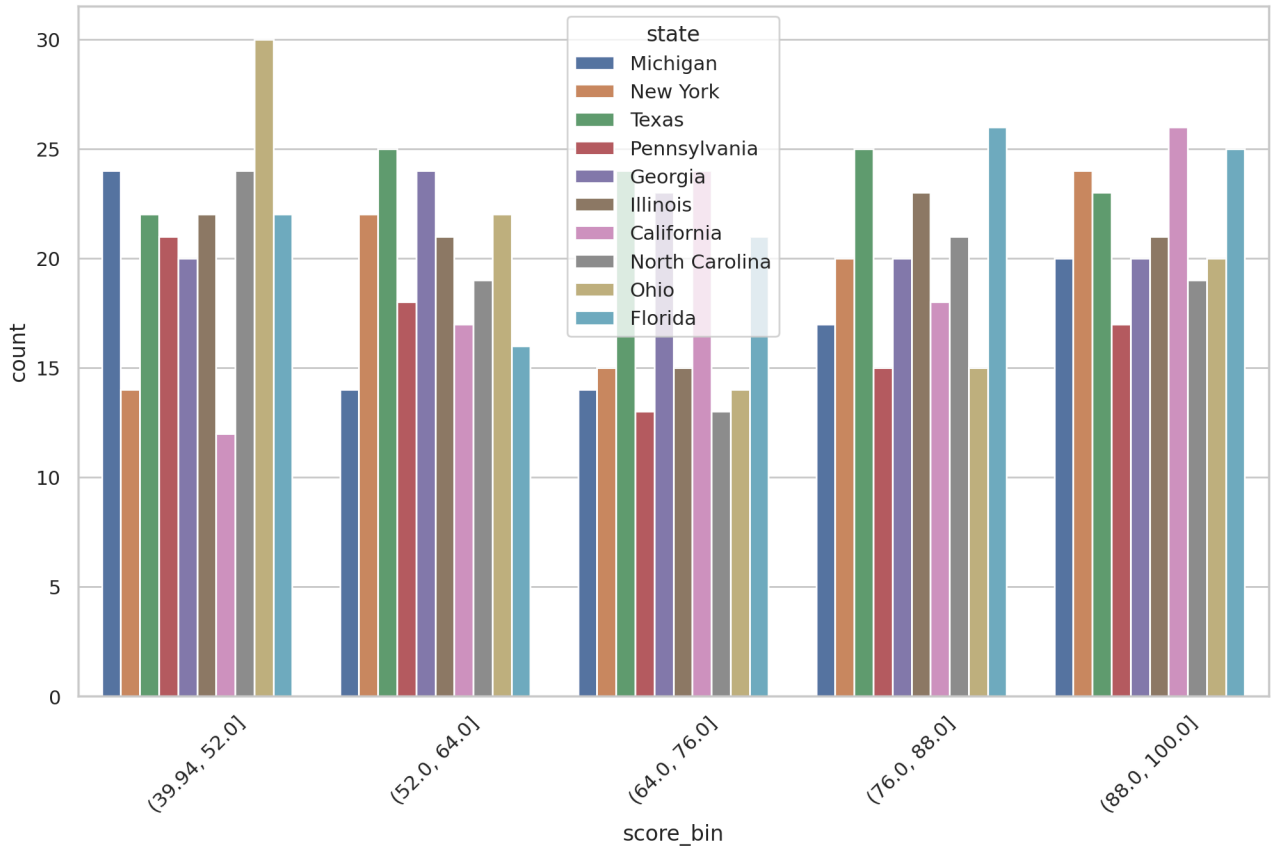
In [ ]:

```
1
```

In [98]:

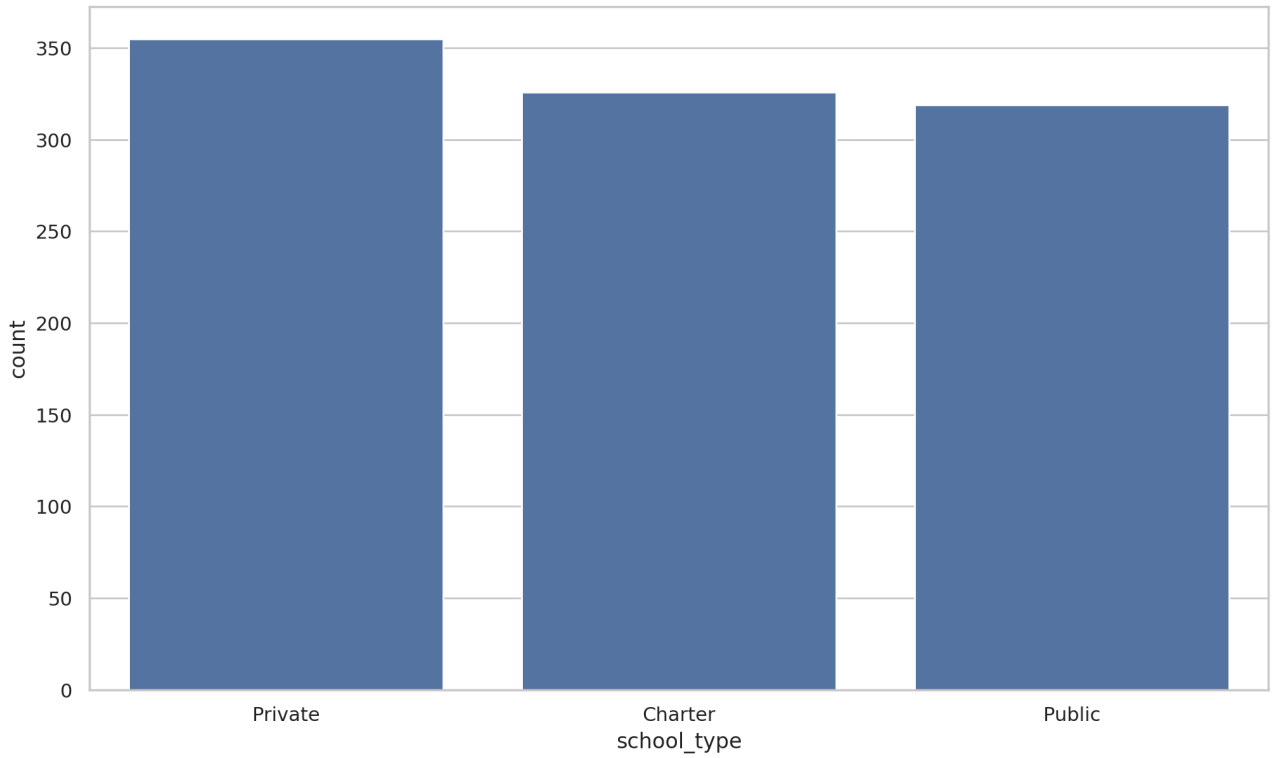
```
education_data_df['average_test_score'] =
pd.cut(education_data_df['avg_test_score_percent'], bins=5)

sns.countplot(x='score_bin', hue='state', data=education_data_df)
plt.xticks(rotation=45)
plt.show()
```

Out[98]:



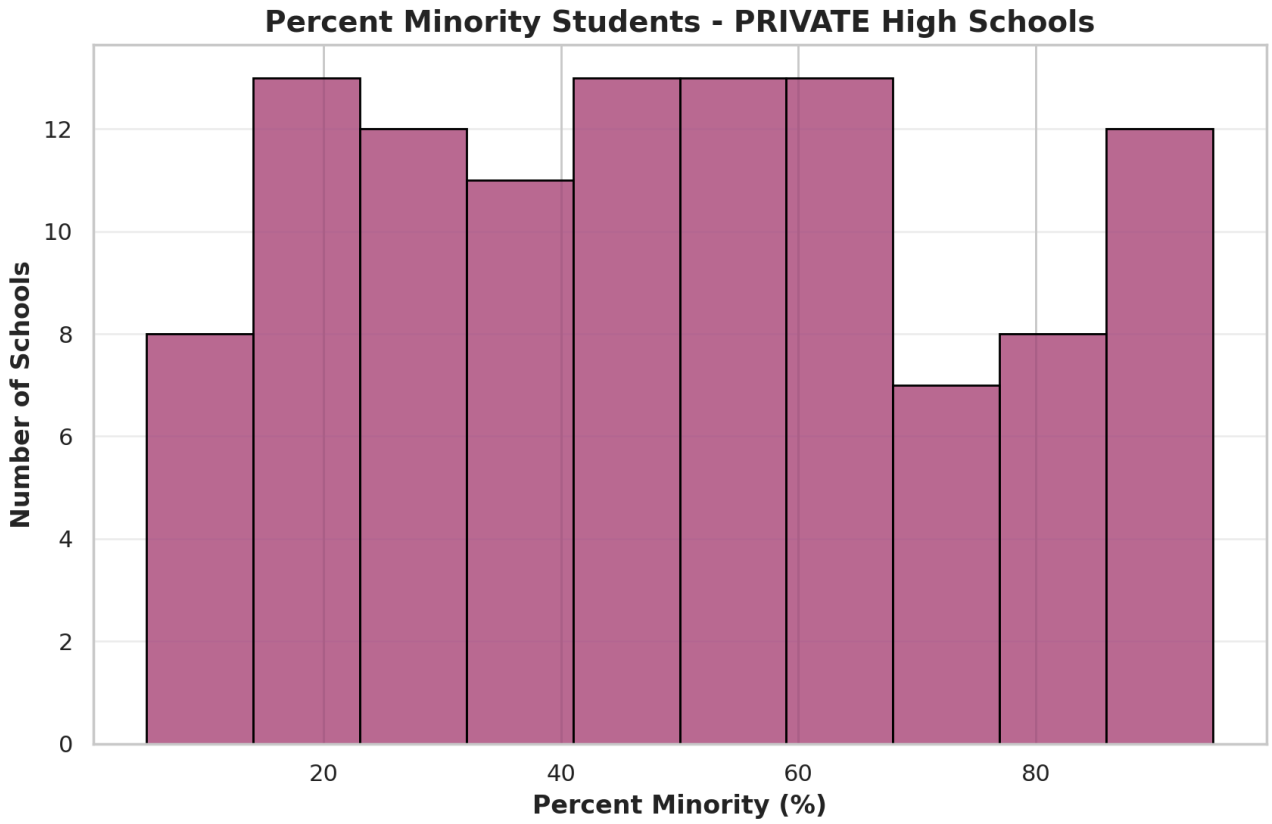

```
sns.countplot(x='school_type', data=education_data_df)
plt.show()
```

Out[96]:




```
private_high = education_data_df[(education_data_df['school_type'] ==
'Private') &
                                 (education_data_df['grade_level'] ==
'High')]

plt.figure(figsize=(10, 6))
sns.histplot(data=private_high,
             x='percent_minority',
             bins=10,
             color='#A23B72',  # Pink
             edgecolor='black',
             stat='count')

plt.xlabel('Percent Minority (%)', fontsize=12, fontweight='bold')
plt.ylabel('Number of Schools', fontsize=12, fontweight='bold')
plt.title('Percent Minority Students - PRIVATE High Schools',
fontsize=14, fontweight='bold')
plt.grid(axis='y', alpha=0.3)
```

Out[102]:


Percent Minority Students - PRIVATE High Schools
Number of Schools
0
2
4
6
8
10
12
20
40
60
80
Percent Minority (%)